

Does the machinery transfer? WS1 and WS2 on machine 3

Phase 2, first item — the same two theorems in base 3, and what the constants say — July 26, 2026

1. Why this was the test worth running

Phase 1 produced two results about the Space Needle and a pile of machinery. The program's fourth goal is a toolkit that *transfers*, and the honest way to test that is to point the machinery at a machine it was not designed for — a different base, a different halting set, a different origin. Machine 3 qualifies: it is genuinely two-variable, its halting set is the powers of 27, and it lives in base 3.

The enabling observation is that machine 3 has the same shape. Its own T4 says $b = 1$ at every reset, so the a -values at resets follow a one-variable map, and writing $a = 3^j(3m + r)$ with r in $\{1, 2\}$ that map is

$$G(a) = a + m + j + c_r = (3^{j+1} + 1)m + (r \cdot 3^j + j + c_r), \quad c_1 = 3, c_2 = 4$$

— the exact base-3 analogue of the Needle's $(2^{v+1}+3)k + (2^v+v)$, verified against machine 3's own accelerated step on 59,988 values. Both machines are **branch-affine**: in base q , each valuation branch is an affine map of the tail. That is the property the whole Phase-1 apparatus actually needs, and naming it is the transferable part.

2. WS1 in base 3

states n	structures searched	transitions	certificate for machine 3
1	1	3	no certificate
2	56	6	no certificate
3	7,965	9	no certificate
4	2,128,064	12	no certificate

So: **machine 3 has no 3-automatic non-halting certificate with at most 4 states**, by the same exhaustive, exactly-decided search. The search code is shared — the base- q version reproduces the Needle's pair sets and refutations identically, which is how it is validated — and machine 3 enters it as nothing but a branch table.

Sizes are not directly comparable across alphabets: a base-3 automaton of n states carries $3n$ transitions against a base-2 automaton's $2n$. Counted in transitions, the Needle is refuted through 14 and machine 3 through 12 — the same neighbourhood, reached at very different enumeration costs (256 million structures against 2.1 million). Nothing here suggests one machine is harder than the other; the honest statement is that both resist certificates of the same small size.

3. WS2 in base 3

The density argument transfers line for line, with only constants changing: expansion $a + 3 \leq G(a) \leq 2a$; at most one preimage per branch, so $d(y) \leq 2\log_3 y + 2$; and completeness — a seed $a \leq x$ halting within

L resets reaches a power of 27 below $2^L x$, so the depth-graded backward enumeration is exact and complete, validated here set-for-set against forward simulation. The theorem becomes $\#\{a \leq x \text{ halting within } L \text{ steps}\} \leq (L+1)(2\log_3 x + 2L + 2)^{L+1}$.

The interesting part is the constant. For each machine the average backward branching has a rigorous ceiling — the sum of $1/A$ over branches, since each branch is one residue class — and in both cases the measurement sits on the ceiling:

machine	branching sum	rigorous ceiling	measured	halting seeds, depth ≤ 5
Space Needle (base 2)	$\sum_v 1/(2^{v+1}+3)$	0.5453	0.5452	558 at 10^{96} , growing like $(\log_2 x)^{1.0}$
Machine 3 (base 3)	$\sum_{j,r} 1/(3^{j+1}+1)$	0.8081	0.8080	136 at 10^{96} , linear in $\log_3 x$

Both are subcritical, which is why the halting basins are thin, and both measurements agree with the ceiling to four decimals. That agreement is itself a finding: the divisibility conditions that decide whether a branch has a preimage behave, in aggregate, exactly as independent events. It is also precisely the assumption the unbounded-depth theorem would need *along the tree* rather than on average — so the two machines now give the same evidence for the same open statement, and the same reason it stays open.

Machine 3 sits closer to criticality (0.81 against 0.54), and its halting basin is correspondingly bushier per root: the depth- ≤ 5 total is 2.0 times the number of powers of 27 below x , against 1.75 times the powers of 2 for the Needle.

4. What this does and does not establish

claim	status
The toolkit transfers (G4)	ESTABLISHED, and more cheaply than expected: machine 3 needed one branch table and no new theory. The reusable abstraction is branch-affine in base q — write $x = q^{ p }m + \text{val}(p)$ and require $F(x) = A_p m + B_p$.
Two more theorems (G1)	ESTABLISHED: no small 3-automatic certificate for machine 3, and its depth-graded density bound with complete exact counts.
A hardness ranking across machines (G3)	NOT ESTABLISHED. Refutation sizes in different bases are not commensurable, and both machines fall in the same narrow band anyway. A real ranking needs the sizes pushed well past exhaustive enumeration.
Any decision (G2)	NOTHING. As in Phase 1, both outcomes are negative-side results. Deciding an instance still depends on WS3 (Baker) or on certificate searches an order of magnitude larger.

Code: `automatic/general.py` (base- q search, validated against the base-2 code), `automatic/machine3_map.py` (branch table, verified against machine 3's accelerated step), `automatic/search_m3.py`, `density/density_m3.py` (lemmas, complete enumeration, forward cross-check).